Activity 3.3 – CPU Self-Check Module

1.Design Approach:

Designed a CPU self-checking support module using Verilog HDL to monitor processor signals and identify invalid program counter alignment conditions. Implemented clock-based monitoring logic.

2.Verification Steps:

- Created a separate testbench.
- Applied reset to initialize the module.
- Provided sample PC and instruction values.
- Tested aligned and misaligned PC conditions.
- Observed the monitoring output.
- Simulated using ModelSim.

3.Results:

The testbench checks the specified program counter alignment condition. PASS/FAIL messages are displayed.

4.Issues Faced:

Processor signals must be connected correctly. Program counter alignment conditions require proper checking. Reset handling must be synchronized.

5.Conclusion:

Successfully designed and verified a CPU self-checking support module using Verilog HDL.